

Machine Learning Lab Projects

Select One Project Per Group

Overview

Each group must select one project from the list below. You are expected to independently find a suitable dataset, apply appropriate Machine Learning techniques, and demonstrate a working system at the end of the semester.

Available Projects

#	Project Title	Brief Description
1	Network Intrusion Detection System	Build a smart system that identifies malicious or suspicious network traffic. Explore both supervised and unsupervised ML techniques, and compare their effectiveness in detecting intrusions.
2	Student Performance Prediction	Predict students final grades using historical academic records, attendance, and personal background data. Apply regression or classification models and analyze which features influence performance the most.
3	Student Dropout Risk Predictor	Develop an early-warning system that flags students likely to drop out before it is too late. Use academic, attendance, and demographic data and ensure your model explains why a student is at risk.
4	CI/CD Build Failure Predictor	Predict whether a software build will fail before it runs, using commit metadata. Expose your model as a REST API that can serve as a pre-build gate in real CI pipelines.
5	Water Potability Classifier	Classify whether a water sample is safe to drink from chemical sensor readings. Identify the most dangerous chemical factors and deploy the model as a public API for real-world use.
6	Hospital Appointment No-Show Predictor	Predict which patients are likely to miss their appointment using demographics, booking info, and reminder data. Identify the strongest behavioral predictors to help design better intervention strategies.
7	Industrial Predictive Maintenance	Predict potential mechanical failures in industrial machinery before they occur using time-series sensor logs. Feature engineering and high recall are key, a missed failure is costly.

#	Project Title	Brief Description
8	Real Estate / Dynamic Pricing Engine	Build a regression model to predict property prices from structural, location, and economic features. Handle skewed distributions, encode categorical variables, and evaluate using RMSE and MAE.
9	Phishing / Spam Detection System	Develop a binary classifier to detect malicious emails or URLs using text and URL features. Apply NLP preprocessing, TF-IDF vectorization, and evaluate with F1-Score.
10	Collaborative Recommender System	Build a movie or book recommendation engine that suggests items based on user ratings. Use matrix factorization or KNN to handle sparse user-item interaction data.

Required Steps (All Projects)

Every group must complete all of the following, regardless of which project they choose:

- Find a dataset
- Apply ML techniques and train a model
- Find the best possible model with hyperparameter tuning.
- Demo the project in the last lab using FastAPI or a web server

Demo Schedule

- Live demos will be held in the second last lab session of the semester.
- Each group will be allotted a fixed time slot for demonstration and viva.
- Submit one printed report per group on the day of your demo.
- No online report submissions will be accepted.

Important Policies

- No extensions will be granted under any circumstances.
- **Plagiarism** or code copying will result in zero marks for the entire group.

Evaluation & Submission Criteria

Marks Distribution

Component	Marks
Live Demo + Viva	20
Printed Report	10
Total	30